

KARTHIK KUMAR VASUDEVA

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PROFILE

Applied and Computational Mathematics PhD Candidate with expertise in C++, numerical algorithms, and high-performance computing. Accomplished educator in applied and computational mathematics, specializing in curriculum development and innovative teaching strategies. Designed and implemented engaging courses in programming and mathematical analysis, enhancing student comprehension and critical thinking. Proven record in research in finite element methods, optimizing computational efficiency, and mentoring diverse learners. Committed to fostering an inclusive, research-driven academic environment that promotes both student success and the practical application of mathematical concepts to real-world problems.

EDUCATION

Doctor of Philosophy in Applied Mathematics *August 2021 - Present*

Doctoral Candidate

Expected Graduation: July 2026

Texas Tech University, Texas, USA

Master of Science in Applied and Computational Mathematics *January 2019 - May 2021*

Texas A&M University-Corpus Christi, Texas, USA

Bachelor of Science in Civil Engineering *August 2014 - June 2018*

Visvesvaraya Technological University, Karnataka, India

RESEARCH

Doctor of Philosophy *August 2021 - Present*

- Advisors: Dr. Eugenio Aulisa (Chair) and Dr. Giorgio Bornia (Co-Chair)
- Tentative Dissertation Title: Finite Element Methods for Boundary Control Problems Constrained by the Mixed Biharmonic Equation.
 - Develop mixed finite element discretizations for a class of optimal control problems constrained by higher order PDEs.
 - Analyze well-posedness and derive error estimates in Sobolev spaces.
 - Implement scalable solvers in C++ for large sparse saddle-point systems

Master's Thesis *January 2020 - May 2021*

Texas A&M University-Corpus Christi, Texas, USA

- Thesis Title: On a Finite Element Model for Two-Dimensional Thermoelastic Problems.
- Research Advisor: Dr. Mallikarjunaiah S. Muddamallappa Department of Mathematics and Statistics - Texas A&M University-Corpus Christi.

Undergraduate Research *August 2017 - June 2018*

The Oxford College of Engineering, India

- Project Title: Odour Regulation - A Solution to Control Odor from Municipal Solid Waste Management Plant.

- Research Advisor: Prof. Amruth Kanakaraj
Department of Civil Engineering, The Oxford College of Engineering.

RESEARCH INTERESTS

Applied and Computational Mathematics, Finite Element Methods, Optimal Control, Computational Mechanics, Partial Differential Equations, Scientific Computing.

PUBLICATIONS

1. Distributed Control Approach for Boundary Control Problem Constrained by the Biharmonic Equation. Manuscript currently in preparation for submission to a peer-reviewed journal.
2. Mixed Finite Element Methods for Boundary Control Problems Constrained by the Biharmonic Equation. In preparation. Abstract accepted, 15th AIMS Conference, Athens, Greece, 2026.
3. Generalized Mathematical Models for Unoccupied aerial vehicles in a simulated environment. Manuscript currently in preparation for submission to a peer-reviewed journal.
4. SwarmFly: A comprehensive simulation tool for UAV swarm experiment design and validation. Submitted to a peer-reviewed journal.
5. Hyun Chul Yoon, Karthik K Vasudeva, and SM Mallikarjunaiah. "Finite element model for a coupled thermo-mechanical system in nonlinear strain-limiting thermoelastic body". In: *Communications in Nonlinear Science and Numerical Simulation* (2022), p. 106262
6. Phadke, A, C N Sekharan, Karthik K Vasudeva, D Kar, "Unifying Sky and Chain: A Perspective on Blockchain's Role in UAV Swarms". In: 2024 2nd International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings). IEEE. 2024, pp. 1–9.

PROFESSIONAL EXPERIENCE

Please Click [here](#) to view Teaching Portfolio, Teaching Evaluations and Student Projects.

Graduate Part-Time Instructor (Instructor of Record)

January 2025 - Present

Department of Mathematics and Statistics, Texas Tech University

Fully responsible for course design and instruction, including syllabus development, lecture delivery, assessment creation, and grading for large undergraduate classes (typically 90–120 students per term). Designed and administered exams, quizzes, and programming-based assignments emphasizing mathematical rigor, algorithmic thinking, and computational problem-solving.

Summer 2026: Calculus III, Math 2450-D02.

Spring 2026: Calculus II, Math 1452.

Fall 2025: Introductory Mathematical Analysis I, Math 1330-113.

Summer 2025: Linear Algebra, Math 2360-D11.

Spring 2025: Calculus I, Math 1451-113.

Research Assistant

August 2024 - December 2024

Department of Mathematics and Statistics, Texas Tech University

Studied and investigated various mixed finite element method discretizations to solve the Biharmonic Equation with clamped boundary conditions. Collaborated with faculty and peers to analyse results and refine methodologies. Contributed to academic papers and presentations.

Teaching Faculty

May – July, 2024 – 2026

TexPREP-Lubbock

Course and Lead Instructor: Fully responsible for the end-to-end design, development, and delivery of curriculum for distinct computer science tracks.

Introduction to C++ Programming, Summer 2024, 2025, 2026

Programming with JavaScript II

Programming with JavaScript I, Summer 2024, 2025, 2026

Introduction to Programming with Python, Summer 2026

Graduate Part-Time Instructor (Instructor of Record)

August 2022 - July 2024

Department of Mathematics and Statistics, Texas Tech University

- Summer 2024: Calculus 1, Math 1451-D21.
- Spring 2024: Calculus 1, Math 1451-021.
- Fall 2023: College Algebra, Math 1320-112.
- Summer 2023: Introductory Mathematical Analysis I, Math 1330-D21.
- Spring 2023: Introductory Mathematical Analysis II, Math 1331-007.
- Fall 2022: Introductory Mathematical Analysis I, Math 1330-112.

Tutor

August 2021 – Present

Texas Tech University, Lubbock, TX

Provide individual and group tutoring in Analytical Geometry, Calculus I–III, College Algebra, Computational Techniques for Science and Mathematics, Contemporary Math, Differential Equations I–II, Intro to Math Analysis I–II, Trigonometry, Statics, Mathematical Computing, Physics, PreCalculus, Linear Algebra.

Graduate Teaching Assistant

August 2021 - July 2022

Department of Mathematics and Statistics, Texas Tech University

Assisted the lecture instructor in delivering the Intro to Mathematical Analysis (Math-1330, Math-1331) and Analytical Geometry courses.

- Wrote multiple-choice questions (MCQs) for the test bank.
- Graded assignments and exams for the Analytic Geometry (Math-1350) course.

Graduate Teaching Assistant (Instructor of Record)

January 2019 - May 2021

Department of Mathematics and Statistics, Texas Tech University

- Calculus III (Math-2415), Spring 2021
- Calculus II (Math-2414), Fall 2020
- Calculus I (Math-2413), Spring 2020, Summer 2020
- Statistics for Life (Math-1442), Spring 2019, Summer 2019, Fall 2019

SKILLS AND INTERESTS

Programming Languages and Applications

C++, MATLAB, L^AT_EX, Python, JavaScript & p5.js, LMS, AutoCAD, Staad Pro, Etabs, Google Classroom, IXL, Google Workspace tools.

Mathematical Software

Finite Element software libraries, Sparse Linear Solvers (Saddle-point), Git GitHub ParaView (Scientific Visualization & Post-Processing).

Operating Systems

Windows, Linux.

Hardware & Environment

Linux / Unix Shell Scripting, Multi-threaded Optimization (OpenMP), Distributed Computing Concepts (MPI), Numerical Linear Algebra, Sparse Matrix Data Structures.

Personal

Creating Educational video content, teaching innovations, tutoring math and engineering courses, chess, hiking...

Languages Known

English (speaking, writing, reading technical and science documentation), Telugu, Tamizh, Kannada (Native tongue)

PROJECTS

- **Boundary Optimal Control for Biharmonic Equations** Developed mixed finite element solvers in **C++** for boundary optimal control problems constrained by the biharmonic equation using Hermann–Miyoshi and Ciarlet–Raviart discretizations, block-structured saddle-point systems. Implemented lifting-function techniques to handle Dirichlet boundary controls, improving numerical stability, optimal convergence and efficient assembly of large sparse matrices for high-order PDE-constrained optimization and scientific computing. Currently working on analyzing the well-posedness in integer order Sobolev Spaces, error estimates and convergence rates. Fall 2025.
- **Distributed Optimal Control using Finite Element Methods** Implemented distributed optimal control solvers for elliptic and biharmonic PDEs in **C++**, including error estimation and convergence studies. The project strengthened skills in large-scale sparse linear systems, variational formulations, and performance-aware scientific programming. Spring 2025.
- **Mixed Finite Element Methods for Higher-Order PDEs** Designed and implemented mixed finite element formulations for fourth-order PDEs using **C++**, reducing regularity requirements and enabling efficient C^0 discretizations. Gained experience in block-matrix structures, inf-sup stability, and scalable solver design, exhibiting the connections between variational formulations, numerical linear algebra and high performance PDE solvers. Fall 2024.
- **Neural Network Solvers for Differential Equations** Implemented feed-forward neural networks in **Python** to approximate solutions of first- and second-order ordinary differential equations. Explored loss-based physics-informed training, gaining experience at the intersection of numerical analysis and machine learning including GPU-accelerated optimization, automatic differentiation and the integration of classical solvers with ML workflows. Spring 2023.
- **Parallel QR Factorization with Partial and Full Pivoting** Implemented parallel QR Factorization algorithm with partial and full pivoting in **Python** using shared-memory parallelism. Analyzed numerical stability, flops and computation cost. Gained hands-on experience with numerical linear algebra, data locality, and parallel scalability concepts directly relevant to GPU-accelerated solvers and HPC architectures. Fall 2022.
- **Programming Instruction for Computational STEM Applications**
Designed and taught hands-on programming modules in **C++** and **JavaScript** for STEM students, with an emphasis on algorithmic thinking, numerical logic, and mathematical problem solving. Student projects in C++ included Morse-code translators, blackjack simulations, currency converters, prime number analyzers, and interactive text-based games, reinforcing control flow, data structures, and computational efficiency. JavaScript projects using the **p5.js** library included animated scenes, GIF-based storytelling, kaleidoscope visualizations, and numerical fractals such as the **Mandelbrot set**, car racing timer introducing concepts of iteration, and visualization of mathematical phenomena. This experience strengthened the connection between mathematical modeling, algorithm design, and scalable computation relevant to scientific computing and

simulation workflows. With TexPrep.

PRESENTATIONS AND TALKS

1. "Boundary Optimal Control Problem Constrained by the Fourth-Order Elliptic PDE", Annual Cross-Disciplinary Faculty Conference, Texas Tech University, April, 2026.
2. "Mixed Finite Element method for Optimal Control Problems Constrained by the Biharmonic Equation", Graduate Student Research Day, SIAM chapter of Texas Tech University, Texas, March, 2026.
3. "Distributed Control approach for Boundary Control Problems Constrained by the Biharmonic Equation", 25th Annual Graduate Poster Competition, Texas Tech University, March, 2026.
4. "Mixed Finite Element method for Boundary Control Problems Constrained by the Biharmonic Equation", Research at the Interface of Applied Mathematics and Machine Learning, CBMS Conference, University of Houston, December, 2025.
5. "Finite Element Methods for Optimal Control Problems Constrained by the Biharmonic Equation, Qualifying Exam, Texas Tech University, Oct 2025.
6. "On a Finite Element Model for Two-Dimensional Thermoelastic Problem", Graduate Student Research Day, SIAM chapter of Texas Tech University, Texas, March, 2022.
7. "On a Finite Element Model for Implicit Thermoelasticity" at The 5th Coastal Bend Mathematics and Statistics Conference, Texas A&M University- San Antonio, 2021.
8. Karthik Kumar et al. "Odour regulation- A solution to control odour from municipal solid waste management plant". In: *International Journal of Creative Research Thoughts* 6 (2 Apr. 2018), pp. 60–66, 2018.

CONFERENCES AND WORKSHOPS

1. Geometric Analysis and Applications, Red Raider Minisymposia, Texas Tech University, April 25, 2026
2. CBMS Conference-Research at the Interface of Applied Mathematics and Machine Learning, University of Houston, December 8–12, 2025.
3. Differential Geometry and Integrable Systems, Texas Tech University, Texas, April 20-23, 2023.
4. Texas Differential Equations Conference, Texas A&M University - Corpus Christi, Texas, March 30-31, 2019.
5. Karthik Kumar et al. "Odour regulation- A solution to control odour from municipal solid waste management plant". In: *International Journal of Creative Research Thoughts* 6 (2 Apr. 2018), pp. 60–66, 2018.
6. Workshop on "Seismic Performance Assessment of Structures" by the Department of Civil Engineering, Alpha College of Engineering, Karnataka, India, 2018.
7. A 2-day project-based training program on Bridge Design Conducted by Civil Simplified at JSS Academy of Technical Education, Karnakata, India, 2018.
8. A 2-day project-based training program on seismic design at Skifi labs, Karnataka, India, December 2017.

LEADERSHIP EXPERIENCE

Vice President, SIAM-TTU Student Chapter

June 2024 - May 2025

Society for Industrial and Applied Mathematics (SIAM), Texas Tech University

- Organized and led monthly meetings to plan chapter activities and initiatives. Responsible for hosting a colloquium and representing the chapter at university-wide events.
- Founder of SIAM Math Quiz.

SERVICES

Mentor

March 2026 - Present

Math Circles, Department of Mathematics and Statistics, Texas Tech University

Judge

February 2026

70th Annual South Plains Regional Science and Engineering Fair

Judge

2022, 2026

FIRST Robotics Competition

- Evaluated high school robotics projects based on engineering design, technical execution, and teamwork; provided constructive feedback to encourage innovation and problem-solving.

Judge

December 2025

Harmony Science Academy Science Fair, Harmony Science Academy, Lubbock, Texas

- Served as a judge for an annual K–8 science and engineering fair, evaluating student research projects across science, technology, engineering, art, and mathematics (STEAM).

Volunteer

May 2024, 2025

Emmy Noether Day, Department of Mathematics and Statistics, Texas Tech University

- Supported departmental outreach efforts through event setup, coordination, and interaction with students and faculty during mathematics-focused activities.

AWARDS AND ACHIEVEMENTS

SIAM Teaching Award,

Texas Tech University

April 2026.

SIAM Graduate Research Day Award,

Texas Tech University

April 2026.

CBMS Conference Travel Grant,

University of Houston

December 2025.

Texas Tech SIAM Scholarship Award,

Texas Tech University

August 2025.

Thomas McLaughlin Scholarship,

Texas Tech University

August 2024.

Gordon Fuller Mathematics Scholarship Endowment,

Texas Tech University

August 2023.

International Presidential Graduate Scholarship,

Texas A&M University - Corpus Christi

January 2021.

*College of Science & Engineering Graduate Scholarship,
Texas A&M University - Corpus Christi*

August 2020.

*College of Science & Engineering Graduate Scholarship,
Texas A&M university - Corpus Christi*

January 2020.

Outstanding Student Award (Topper in Academics) 4.0 on a scale of 4.0,

Department of Civil Engineering, The Oxford College of Engineering, Karnataka, India. *June 2018.*

Best Research Paper Award for “Odour Regulation- A solution to control odor from municipal solid waste management plant” (Undergraduate degree). National Conference on Science, Engineering, and Management, The Oxford College of Engineering, Karnataka, India.

April 2018.

PROFESSIONAL MEMBERSHIP

- American Mathematical Society.
- Society for Industrial and Applied Mathematics.

REFERENCES

Research References:

Dr. Eugenio Aulisa

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Dr. Giorgio Bornia

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Dr. Mallikarjunaiah S. Muddamallappa

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Teaching References:

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Dr. Brock Williams

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GOOGLE SCHOLAR



LINKEDIN



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